Exercise C - II

1. With usual notation, if in a Δ ABC, $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$; then prove that, $\frac{\cos A}{7}=\frac{\cos B}{19}=\frac{\cos C}{25}$.

एक Δ ABC में यदि $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$ है तो सिद्ध कीजिए $\frac{\cos A}{7}=\frac{\cos B}{19}=\frac{\cos C}{25}$

Ans.

Sol. $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}=k$ (let)

$b + c = 11k$ $\quad$ $a = 18k - 11k$

$a + c = 12k$ $\quad$ $a = 7k$

$a + b = 13k$ $\quad$ $b = 6k$

$(a + b + c) = 18k$ $\quad$ $c = 5k$

$$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{36k^2+25k^2-49k^2}{2\times 6k\times 5k}$$

$$\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{49k^2+25k^2-36k^2}{2\times 7k\times 5k}$$

$$\cos C=\frac{a^2+c^2-b^2}{2ab}=\frac{49k^2+36k^2-25k^2}{2\times 7k\times 6k}$$

2. Given a triangle ABC with AB = 2 and AC = 1. Internal bisector of $\angle$BAC intersects BC at D. If AD = BD and Δ is the area of triangle ABC, then find the value of $12\Delta^2$.

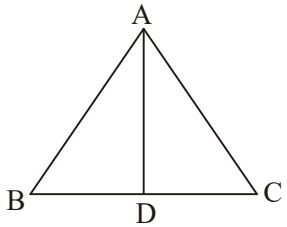
दिये गये त्रिभुज ABC में AB = 2 एवं AC = 1 है। कोण $\angle$BAC का कोणार्द्धक BC को D पर प्रतिच्छेद करता है। यदि AD = BD तथा त्रिभुज ABC का क्षेत्रफल Δ है तो $12\Delta^2$ का मान ज्ञात कीजिये।

Ans. 9

Sol. AB = 2, aC = 1, BC = a

$$CD=\frac{1}{1+2}a=\frac{a}{3}$$

$$BD=\frac{2a}{3}$$



length of bisector

$$\frac{\left(2bc\cos\frac{A}{2}\right)}{b+c}=\frac{2a}{3}$$

$$\frac{2\times 2\times 1}{3}\cos\frac{A}{2}=\frac{2a}{3}\Rightarrow\cos\frac{A}{2}=\frac{a}{2}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5 - a^2}{3}$$

$a^2 = 3$

$$12D^2 = 12 \times \left(\frac{1}{2} bc \sin A\right)^2$$

$= 9$

3. In a triangle ABC, let angles A, B, C are in G.P. with common ratio 2. If circumradius of triangle ABC is 2, then find the value of $(b^{-1} + c^{-1} - a^{-1})$.

त्रिभुज ABC में, कोण A, B, C गु.श्रे. में है जिसका सर्वानुपात 2 है। यदि त्रिभुज ABC के परिवृत की त्रिज्या 2 है, तो $(b^{-1} + c^{-1} - a^{-1})$ का मान ज्ञात कीजिये।

Ans. 0

Sol. $A = a$ $\qquad R = 2$

$B = 2a$ $\qquad A + B + c = \pi$

$C = 4a$ $\qquad a = \frac{\pi}{7}$

$$\angle A = \frac{\pi}{7} \ \& \ \angle B = \frac{2\pi}{7} \ \& \ \angle C = \frac{9\pi}{7}$$

$$\frac{1}{b} + \frac{1}{c} - \frac{1}{a} \qquad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\frac{1}{2R}\left[\frac{1}{\sin B} + \frac{1}{\sin C} - \frac{1}{\sin A}\right]$$

$$\frac{\sin B + \sin C}{\sin B \sin C} \to \frac{\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{\sin B \sin C}$$

$$\Rightarrow \frac{2\sin\frac{3\pi}{7}\cos\left(\frac{\pi}{7}\right)}{\sin\left(\frac{2\pi}{7}\right)\sin\left(\frac{4\pi}{7}\right)} = \frac{\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}\sin\frac{4\pi}{7}} \Rightarrow \frac{1}{\sin\frac{\pi}{7}}$$

$$\frac{1}{2R}\left(\frac{1}{\sin\frac{\pi}{7}} - \frac{1}{\sin\frac{\pi}{7}}\right) = 0$$

4. For any triangle ABC, if $B = 3C$, show that $\cos C = \sqrt{\frac{b+c}{4c}}$ and $\sin\frac{A}{2} = \frac{b-c}{2c}$.

एक ΔABC में यदि $B = 3C$ है तो दर्शाइए $\cos C = \sqrt{\frac{b+c}{4c}}$ and $\sin\frac{A}{2} = \frac{b-c}{2c}$

Ans.

Sol. $B = 3C \Rightarrow \quad \sin B = \sin 3C$

$$\Rightarrow \frac{\sin B}{\sin C} = 3 - 4\sin^2 C = \frac{b}{c}$$

$3 - 4\sin^2 C = \frac{b}{c}$

$\Rightarrow 3 - 4(1 - \cos^2 C) = \frac{b}{c}$

$\Rightarrow 4\cos^2 C = \frac{b+c}{c}$

$\Rightarrow \cos C = \sqrt{\frac{b+c}{4c}}$

$A + B + C = \pi$

$A = \pi - 4C$

$\frac{A}{2} = \frac{\pi}{2} - 2C$

$\sin\frac{A}{2} = \cos 2C$

$= 2\cos^2 C - 1$

$= \frac{2(b+c)}{4c} - 1$

$\sin\frac{A}{2} = \frac{b-c}{2c}$

5. In a triangle ABC, BD is a median. If $l(BD) = \frac{\sqrt{3}}{4} \cdot l(AB)$ and $\angle DBC = \frac{\pi}{2}$. Determine the $\angle ABC$.

एक त्रिभुज ABC में, BD एक माध्यिका है। यदि $l(BD) = \frac{\sqrt{3}}{4} \cdot l(AB)$ एवं $\angle DBC = \frac{\pi}{2}$ तो $\angle ABC$ ज्ञात कीजिए।

Ans. **120°**

Sol. $P(BD) = \frac{\sqrt{3}}{4}c$

$\frac{1}{2}\sqrt{2a^2 + 2c^2 - b^2} = \frac{\sqrt{3}}{4}c$

$2a^2 + 2c^2 - b^2 = \frac{3c^2}{4}$ _____(1)

In ΔDBC $\quad a^2 = \frac{b^2}{4} - \frac{3c^2}{16}$

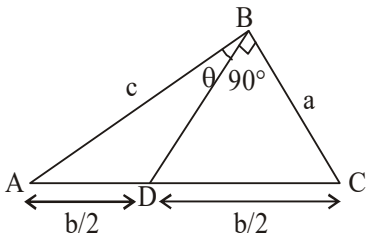
$$2\left(\frac{b^2}{4} - \frac{3c^2}{16}\right) + 2c^2 - b^2 = \frac{3c^2}{4}$$

$$\frac{b^2}{2} - \frac{3c^2}{8} + 2c^2 - b^2 = \frac{3c^2}{4}$$

$$-\frac{b^2}{2} - \frac{3c^2}{8} + 2c^2 - \frac{3c^2}{4} = 0$$

$$-\frac{b^2}{2} + \frac{-3c^2 + 16c^2 - 6c^2}{8} = 0$$

$$-\frac{b^2}{2} + \frac{7c^2}{8} = 0 \Rightarrow 4b^2 = 7c^2$$



$$\cos\theta = \frac{c^2 + \frac{3c^2}{16} - \frac{b^2}{4}}{2 \times c \times \frac{\sqrt{3}c}{4}}$$

$$\cos\theta = \frac{19c^2 - 4b^2}{2 \times 4 \times \sqrt{3}c^2} \Rightarrow \frac{19c^2 - 7c^2}{2 \times 4 \times \sqrt{3}c^2}$$

$$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^o$$

Angle $B = 30^o + 90^o \Rightarrow B = 120^o$

6. In an isosceles ΔABC, if the altitudes intersect on the inscribed circle then find the secant of the vertical angle 'A'.

एक समद्विबाहु ΔABC में यदि सभी लम्ब अन्तःवृत पर प्रतिच्छेद करते है तो शीर्ष कोण 'A' का sec A का मान ज्ञात कीजिये।

Ans. **9**

Sol. $OD = 2r$

$OA = 2R \cos A$

$OA + OD = AD$

$$2r + 2R\cos A = \frac{a}{2}\cot\frac{A}{2}$$

$$2\left[4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right] + 2R\cos A = \frac{2R\sin A}{2}\cot\frac{A}{2}$$

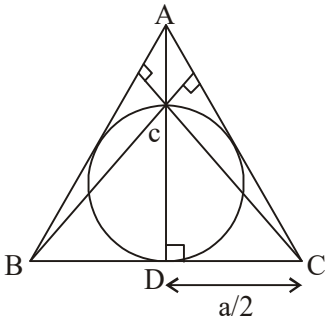
$$\Rightarrow 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}+2\cos A=2\sin\frac{A}{2}\cos\frac{A}{2}\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}}$$

$$8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}+2\cos A=2\cos^2\frac{A}{2}-1+1$$

$$\cos A=1-8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

Now B = C $\qquad$ A + B + C = π

$$C=B=\frac{\pi}{2}-\frac{A}{2}$$



$$\cos A=1-8\sin\frac{B}{2}\sin^2\left(\frac{\pi}{4}-\frac{A}{4}\right)$$

$$1+2\sin^2\frac{A}{2}=1+\frac{8\sin\frac{A}{2}}{2}\left(\sin\frac{A}{4}-\cos\frac{A}{4}\right)^2$$

$$\sin\frac{A}{2}=2\left[1-2\sin\frac{A}{4}\cos\frac{A}{4}\right]$$

$$\sin\frac{A}{2}=2-2\sin\frac{A}{2}$$

$$\sin\frac{A}{2}=\frac{2}{3}$$

$$\cos A=1-2\sin^2\frac{A}{2}=1-2\times\frac{4}{9}=\frac{1}{9}$$

sec A = 9

7. Find the angles of a triangle in which the altitude and a median drawn from the same vertex divide the angle at that vertex into 3 equal parts.

एक त्रिभुज के कोण ज्ञात कीजिए, जिसमें एक शीर्ष से लम्ब एवं मध्यिकाऐं उसके कोण को तीन बराबर भागों में विभाजित करते है।

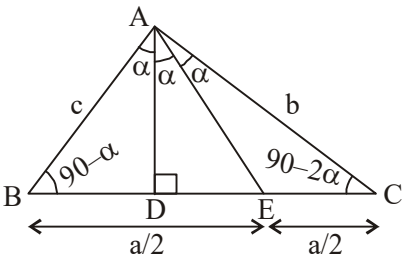
Ans. 30º, 60º and 90º

Sol. A = 3α

In ΔABE

$$\frac{\frac{a}{2}}{\sin 2\alpha}=\frac{AE}{\sin(90-\alpha)}$$

$$\frac{a}{2\sin 2\alpha}\cos\alpha = AE \rightarrow AE = \frac{a}{4\sin\alpha} \qquad \text{_____(1)}$$



In ΔAEC

$$\frac{a}{2\sin\alpha} = \frac{AE}{\sin(90-2\alpha)}$$

$$AE = \frac{a}{2}\frac{\cos 2\alpha}{\sin\alpha} \qquad \text{_____(2)}$$

from (1) & (2), we get

$$\frac{a}{2}\frac{\cos 2\alpha}{\sin\alpha} = \frac{a}{4\sin\alpha} \Rightarrow \cos 2\alpha = \frac{1}{2}$$

$$2\alpha = \frac{\pi}{3}$$

$$\alpha = \frac{\pi}{6}$$

$$A = 3\alpha = 3\left(\frac{\pi}{6}\right) = \frac{\pi}{2}$$

$B = 90 - 30 = 60^\circ$

$C = 90 - 60 = 30^\circ$

8. The ratios of the lengths of the sides BC and AC of ΔABC to the radius of circumscribed circle are equal to 2 and $\frac{3}{2}$ respectively. If the ratio of the lengths of the bisectors of the interior angles B and C is $\frac{\alpha\left(\sqrt{\alpha}-1\right)}{\beta\sqrt{\gamma}}$ where α, β, $\gamma \in N$, then find the value of $(\alpha + \beta + \gamma)$.

Δ ABC की भुजाओं BC तथा AC के लम्बाईयों का परिवृत की त्रिज्या से अनुपात क्रमशः 2 तथा $\frac{3}{2}$ है। यदि कोण B तथा C के कोणार्द्धको की लम्बाई का अनुपात $\frac{\alpha\left(\sqrt{\alpha}-1\right)}{\beta\sqrt{\gamma}}$ है, जहाँ $\alpha, \beta, \gamma \in N$ है। तो $(\alpha + \beta + \gamma)$ का मान ज्ञात कीजिये।

Ans. 18

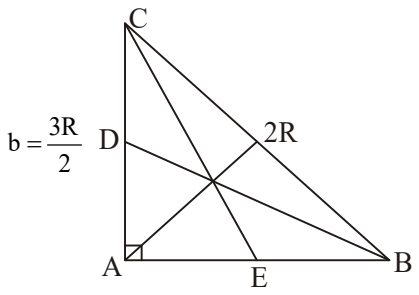
Sol. BC = a & AC = b

$$\frac{a}{R} = 2 \ \& \ \frac{b}{R} = \frac{3}{2}$$

$2\sin A = 2$

$$\sin A = 1 \Rightarrow A = \frac{\pi}{2}$$

$$\sin B = \frac{3}{4}$$



$BD = \sqrt{AD^2 + AB^2}$ $\qquad$ $c = \sqrt{a^2 - b^2}$

$AD = AC\cdot\left(\frac{AB}{AB + BC}\right) = \frac{bc}{a + c}$ $\qquad$ $c = \frac{\sqrt{7}}{2}R$

$AE = \frac{bc}{a + b}$

$CE = \sqrt{AC^2 + AE^2}$

puttin the values of a, b & c, we get

$AD = \frac{3\sqrt{7}R}{2\left(4 + \sqrt{7}\right)}$, $AE = \frac{3\sqrt{7}R}{14}$

$BD = \frac{2\sqrt{7}R}{\sqrt{7} + 1}$, $CE = \frac{3\sqrt{2}R}{\sqrt{7}}$

$\Rightarrow \frac{BD}{CE} = \frac{7\left(\sqrt{7} - 1\right)}{9\sqrt{2}}$

$\alpha = 7, \beta = 9, \gamma = 2$

$\alpha + \beta + \gamma = 18$

9. ABCD is a rhombus. The circumradii of Δ ABD and Δ ACD are 12.5 and 25 respectively. Find the area of rhombus.

ABCD एक समान्तर समबाहु चतुर्भुज है। Δ ABD तथा Δ ACD के परिवृत की त्रिज्या 12.5 तथा 25 क्रमशः है। चतुर्भुज का क्षेत्रफल निकालिए।

Ans. 400 sq. unit

Sol. Let BD = p

AC = q

In ΔABD $\qquad$ In ΔACD

$\frac{q}{\sin\theta_1} = 2R$ $\qquad$ $\frac{q}{\sin\theta_2} = 50$

$p = 2R\sin\theta_1$ $\qquad$ $q = 50\sin\theta_2$

$p = 2\left(\frac{25}{2}\right)\sin\theta_1$

$p = 25\sin\theta_1$

In ΔAMD

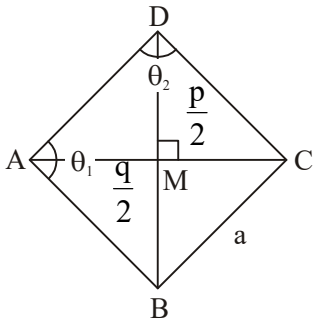
$\frac{\theta_1}{2} + \frac{\theta_2}{2} = \frac{\pi}{2}$

$$\& \ \tan\frac{\theta_1}{2} = \frac{p}{q}$$

$$\text{Area} = \frac{pq}{2}$$

$$= \frac{25\times 50}{2}\sin^2\theta_1 \qquad \tan\frac{\theta_1}{2} = \frac{25\sin\theta_1}{50\sin(\pi-\theta_1)}$$

$$\sin\theta_1 = \frac{2\tan\frac{\theta_1}{2}}{1+\tan^2\frac{\theta_1}{2}} \qquad \tan\frac{\theta_1}{2} = \frac{1}{2}$$



$$\sin\theta_1 = \frac{2\times\frac{1}{2}}{1+\frac{1}{4}} = \frac{4}{5}$$

$$\text{Area} = \frac{25\times 50}{2}\times\frac{16}{25} = 400$$

10. In a triangle ABC if $a^2 + b^2 = 101c^2$ then find the value of $\dfrac{\cot C}{\cot A + \cot B}$.

Δ ABC यदि $a^2 + b^2 = 101c^2$ तो $\dfrac{\cot C}{\cot A + \cot B}$ का मान ज्ञात कीजिए।

Ans. **50**

Sol. $a^2 + b^2 = 101c^2$

$$a^2 + b^2 - c^2 = 100c^2 \Rightarrow \frac{a^2+b^2-c^2}{c^2} = 100$$

$$\frac{\cot C}{\cot A + \cot B} \Rightarrow \frac{\cos C}{\sin C}\frac{\sin A\sin B}{[\cos A\sin B + \cos B\sin A]}$$

$$\Rightarrow \frac{\cos C\sin A\sin B}{\sin C\times\sin(A+B)}$$

$$\Rightarrow \frac{\left(\frac{a^2+b^2-c^2}{2ab}\right)\times\frac{ab}{4R^2}}{\frac{c^2}{4R^2}} = \frac{a^2+b^2-c^2}{2c^2}$$

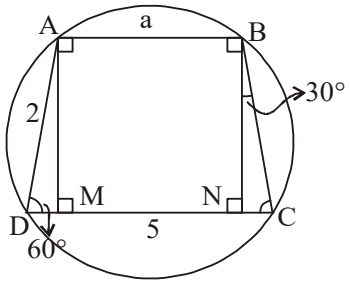
$$\Rightarrow \frac{100}{2} = 50$$

11. The two adjacent sides of a cyclic quadrilateral are 2 & 5 and the angle between them is $60°$. If the area of the quadrilateral is $4\sqrt{3}$, find the remaining two sides.

एक चक्रीय चतुर्भुज में दो क्रमागत भुजाऐं यदि 2 तथा 5 है तथा उनके मध्य कोण $60°$ है। यदि उस चतुर्भुज का क्षेत्रफल $4\sqrt{3}$ है तो बाकी की दो भुजाऐं क्या होगीं।

Ans. 3 cms & 2 cms

Sol. $A = 4\sqrt{3}$



$$A_1 = \frac{1}{2} \times 2\cos 60^\circ \times 2\sin 60^\circ$$

$A_2 = a \times 2 \sin 60^\circ$

$A_3 = A_1$

$AM = 2\sin 60^\circ$

$A_1 + A_2 + A_3 = 4\sqrt{3}$

$BC \cos 30^\circ = BN = AM$

$BC \cos 30^\circ = 2 \sin 60^\circ$

$BC = 2$

$$2 \times \frac{1}{2} \times 2 \times \frac{1}{2} \times 2 \times \frac{\sqrt{3}}{2} + \frac{2a\sqrt{3}}{2} = 4\sqrt{3}$$

$1 + a = 4$

$a = 3$

12. A point 'O' is situated on a circle of radius R and with centre O, another circle of radius $\frac{3R}{2}$ is described. Inside the crescent shaped area intercepted between these circles, a circle of radius R/8 is placed . If the same circle moves in contact with the original circle of radius R, then find the length of the arc described by its centre in moving from one extreme position to the other.

R त्रिज्या वाले वृत्त के किसी बिन्दु 'O' को केन्द्र मानकर एक वृत्त, जिसकी त्रिज्या $\frac{3R}{2}$ है, बनाया जाता है। इन दोनों वृत्तो के बीच बने हुए वलय में R/8 का त्रिज्या का एक वृत्त है। यदि यह वृत्त मूल वृत्त के सम्पर्क में रहते हुए गति करता है तो इसके केन्द्र की गति से निर्धारित चाप की लम्बाई ज्ञात करो जो दो चरम स्थितियों के सापेक्ष बनता है।

Ans. $\frac{7\pi R}{12}$

Sol. Let $\angle ABC = \theta$

B is centre of C_2 then $AB = R$

$$AC = \frac{3R}{2} + \frac{R}{8} = \frac{13R}{8}$$

$$BC = R - \frac{R}{8} = \frac{7R}{8}$$

$$\text{In } \Delta ABC;\ \cos\theta = \frac{AB^2 + BC^2 - AC^2}{2 \times AB \times BC}$$

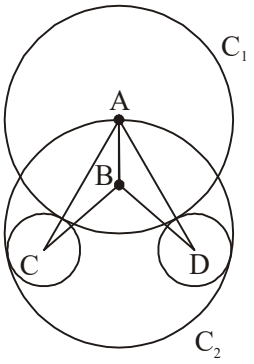
$$\cos\theta = \frac{R^2 + \frac{45R^2}{64} - \frac{169R^2}{64}}{2 \times \frac{7}{8}R^2} = -\frac{56R^2}{64} \times \frac{8}{14R^2}$$

$$\cos\theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

for symmetry $\angle ABD = 120^\circ$

$$\angle BDC = 120^\circ = \frac{2\pi}{3}$$

$$CD = BC \times \frac{2\pi}{3} = \frac{7R}{8} \times \frac{2\pi}{3} = \frac{7\pi r}{12}$$



13. If in a Δ ABC , a = 6, b = 3 and $\cos(A - B) = 4/5$ then find its area.

यदि Δ ABC में a = 6, b = 3 तथा $\cos(A - B) = 4/5$ तो उसका क्षेत्रफल ज्ञात कीजिए।

Ans. 9 sq. unit

Sol. a = 6 & b = 3

$$\cos(A - B) = \frac{4}{5}$$

$$2\cos^2\left(\frac{A - B}{2}\right) = \frac{a}{5}$$

$$\text{Area} = \frac{1}{2} \times 6 \times 3$$

$$\cos\left(\frac{A - B}{2}\right) = \frac{3}{\sqrt{10}} = \frac{A}{K}$$

$$= 9$$

$$\frac{a - b}{a + b}\cot\frac{C}{2} = \tan\left(\frac{A - B}{2}\right) = \frac{1}{3}$$

$$\frac{3}{9}\cot\frac{C}{2} = \frac{1}{3} \Rightarrow C = 90^\circ$$

14. In a Δ ABC

(i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2 \sin A \cos B = \sin C$

(iii) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1 = 0$, prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

Δ ABC में

(i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2 \sin A \cos B = \sin C$

(iii) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1 = 0$ तो सिद्ध कीजिऐ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

Ans.

Sol. (1) $\frac{a}{\cos A} = \frac{b}{\cos B} \Rightarrow a \cos B = b \cos A$

$C = \pi - 2A$

$$a\left(\frac{a^2 + c^2 - b^2}{2ac}\right) = b\left(\frac{b^2 + c^2 - a^2}{2bc}\right)$$

$a^2 + c^2 - b^2 = b^2 + c^2 - a^2$

$a = b \Rightarrow A = B$

(2) $2 \sin A \cos (A) = \sin(\pi - 2A)$

$\sin 2A = \sin 2A$

(3) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1$

$$\Rightarrow \tan^2 \frac{A}{2} + 2 \tan \frac{A}{2}\left(\frac{\pi}{2} - A\right) - 1$$

$$\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \frac{\left(1 - \tan^2 \frac{A}{2}\right)}{2 \tan \frac{A}{2}} - 1 = 0$$

15. Two sides of a triangle are of lengths $\sqrt{6}$ and 4 and the angle opposite to smaller side is 30^0. How many such triangles are possible? Find the length of their third side and area.

एक त्रिभुज में दो भुजाऐं $\sqrt{6}$ एवं 4 सबसे छोटी भुजा के सामने वाला कोण 30^0 है। ऐसे कितने त्रिभुज संभव है ? उसकी तीसरी भुजा की लम्बाई ज्ञात कीजिए।

Ans. two triangle: $(2\sqrt{3} - \sqrt{2})$ & $(2\sqrt{3} + \sqrt{2})$; $(2\sqrt{3} - \sqrt{2})$ & $(2\sqrt{3} + \sqrt{2})$ sq. units

Sol. all sides is $\sqrt{6} < 4$

$$\cos 30^o = \frac{a^2 + 16 - 6}{2a \times 4} = \frac{\sqrt{3}}{2}$$

$$a^2 + 10 = 4\sqrt{3}a$$

$$a^2 - 4\sqrt{3}a + 10 = 0$$

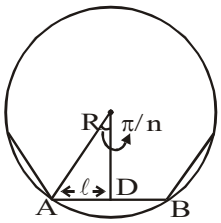
$$a = \frac{4\sqrt{3} \pm \sqrt{48 - 4 \times 1 \times 10}}{2}$$

$$a = \frac{4\sqrt{3} \pm 2\sqrt{2}}{2} \Rightarrow a = 2\sqrt{3} \pm \sqrt{2}$$

16. With reference to a given circle, A_1 and B_1 are the areas of the inscribed and circumscribed regular polygons of n sides, A_2 and B_2 are corresponding quantities for regular polygons of 2n sides. Prove that :
(1) A_2 is a geometric mean between A_1 and B_1
(2) B_2 is a harmonic mean between A_2 and B_1

एक दिए गए वृत में A_1 तथा B_1 उसके n भुजाओं के अन्तः समबहुभुज तथा बाह्य समबहुभुज के क्षेत्रफल है। A_2तथा B_2 सापेक्षतः 2n भुजाओं वाले समबहुभुजों के क्षेत्रफल है। सिद्ध कीजिये :
(1) A_2, A_1 तथा B_1 का गुणोत्तर माध्य है
(2) B_2 , A_2 तथा B_1 का ह. मा. है

Sol.



$\ell = R \cdot \sin \pi/n = AD$

$$A_1 = \frac{n}{2} R^2 \sin \frac{2\pi}{n}$$

$$B_1 = nR^2 \tan \frac{\pi}{n}$$

Replacing n by 2n we get

$$A_2 = nR^2 \sin \frac{\pi}{n}$$

$$B_2 = 2nR^2 \tan \frac{2\pi}{n}$$

$A_1B_1 = n^2R^4\sin^2\frac{\pi}{n} = A_2^2$

Now,

$$\frac{1}{A_2}+\frac{1}{B_1} = \frac{1}{nR^2.\sin\pi/n}+\frac{\cos\pi/n}{nR^2.\sin\pi/n}$$

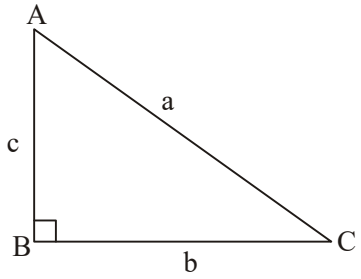
$$= \frac{1}{nR^2.\tan\pi/2n} = \frac{2}{B_2}.$$

17. The triangle ABC is a right angled triangle, right angle at A. The ratio of the radius of the circle circumscribed to the radius of the circle escribed to the hypotenuse is, $\sqrt{2}:\left(\sqrt{3}+\sqrt{2}\right)$. Find the acute angles B & C. Also find the ratio of the two sides of the triangle other than the hypotenuse.

त्रिभुज ABC में कोण A समकोण है। उसमें परिवृत त्रिज्या का एवं कर्ण के बाह्य वृत की त्रिज्या का अनुपात $\sqrt{2}:\left(\sqrt{3}+\sqrt{2}\right)$ है। न्यूनकोण B एवं C ज्ञात करें साथ ही कर्ण के अलावा अन्य दो भुजाओं का अनुपात भी ज्ञात करें।

Ans. $B = \frac{5\pi}{12}; C = \frac{\pi}{12}; \frac{b}{c} = 2+\sqrt{3}$

Sol. $\frac{R}{r} = \frac{\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ (Given)



$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$\frac{r}{R} = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}} = 4\frac{1}{\sqrt{2}}\sin\frac{B}{2}\sin\left(\frac{\pi}{4}-\frac{B}{2}\right)$$

$$\sqrt{3}+\sqrt{2} = \frac{4}{\sqrt{2}}\sin\frac{B}{2}\left(\sin\frac{B}{2}-\cos\frac{B}{2}\right)$$

$$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}} = 2\sin^2\frac{B}{2}-2\sin\frac{B}{2}\cos\frac{B}{2}$$

$$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}} = 1-\cos B-\sin B$$

$$\sin B+\cos B = \frac{\sqrt{2}-\sqrt{3}-\sqrt{2}}{\sqrt{2}} = -\frac{\sqrt{3}}{\sqrt{2}}$$

$$\cos\left(B+\frac{\pi}{4}\right) = -\frac{\sqrt{3}}{2} = \cos\left(\frac{2\pi}{3}\right)$$

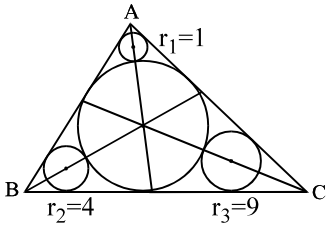
$$B = \frac{2\pi}{3} - \frac{\pi}{4} \Rightarrow \frac{8\pi - 3\pi}{12} = \frac{5\pi}{12}$$

$$C = \frac{\pi}{12}$$

$$\frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow \frac{b}{c} = \frac{\sin\left(\frac{5\pi}{12}\right)}{\sin\left(\frac{\pi}{12}\right)}$$

18. ABC is a triangle. Circles with radii as shown are drawn inside the triangle each touching two sides and the incircle. Find the radius of the incircle of the ΔABC.

त्रिभुज ABC में चित्रानुसार वृत बनाये जाते है जो अन्तः वृत एवं दो भुजाओं को स्पर्श कर रहें है। ΔABC के अन्तः वृत की त्रिज्या ज्ञात कीजिये।



Ans. r = 11

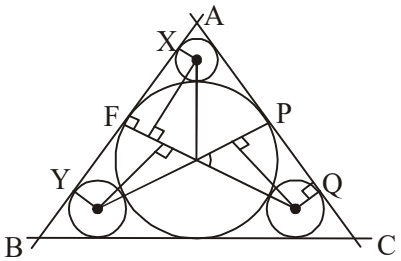
Sol. $\therefore XF = \sqrt{(r+1)^2 - (r-1)^2}$

$2\sqrt{r}$

Similarly

$YF = \sqrt{(r+4)^2 - (r-4)^2} = 4\sqrt{r}$

$PQ = 6\sqrt{r}$



$$\tan\frac{A}{2} = \frac{r-1}{2\sqrt{r}}, \tan\frac{B}{2} = \frac{r-4}{4\sqrt{r}}, \tan\frac{C}{2} = \frac{r-9}{6\sqrt{r}}$$

Using $\sum \tan\frac{A}{2}\tan\frac{B}{2} = 1$

$$\frac{(r-1)}{2\sqrt{r}}\cdot\frac{(r-4)}{4\sqrt{r}} + \frac{r-4}{4\sqrt{r}}\cdot\frac{r-9}{6\sqrt{r}} + \frac{r-9}{6\sqrt{r}}\cdot\frac{r-1}{2\sqrt{r}} = 1$$

Multiplying it throughout by 24r

$3(r-1)(r-4) + (r-9)(r-9) + 2(r-9)(r-1) = 24r$

$6r^2 = 72r + 66 = 0$

$(r-11)(r-1) = 0 \Rightarrow r = 11$ as $r \neq 1$

19. Line l is a tangent to a unit circle S at a point P. Point A and the circle S are on the same side of l, and the distance from A to l is 3. Two tangents from point A intersect line l at the point B and C respectively. Find the value of (PB)(PC).

रेखा l इकाई त्रिज्या के वृत S को P बिन्दु पर स्पर्श करती है। बिन्दु A तथा वृत S रेखा l के एक और है तथा A की l से दूरी 3 है। A से दो स्पर्श रेखाऐं l को B एवं C पर क्रमशः प्रतिच्छेद करती है। (PB)(PC) का मान ज्ञात कीजिये।

Ans. 3

Sol. $AP = 3$ & $AP_1 = 1$

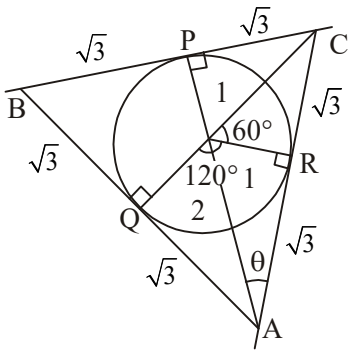
$AP \times AP_1 = AR^2$

$3 \times 1 = AR \rightarrow AR = \sqrt{3}$

$\angle A = 2\theta = 60^\circ$ $\qquad$ $\tan\theta = \dfrac{1}{\sqrt{3}}$

$\theta = 30^\circ$

$(PB)(PC) = \left(\sqrt{3}\right)^2 = 3$



20. Let ABC be an acute triangle with orthocenter H. D, E, F are the feet of the perpendiculars from A, B, and C on the opposite sides. Also R is the circumradius of the triangle ABC. Given $(AH)(BH)(CH) = 3$ and $(AH)^2 + (BH)^2 + (CH)^2 = 7$. Then find :

(a) the ratio $\dfrac{\prod \cos A}{\sum \cos^2 A}$, (b) the product $(HD)(HE)(HF)$ (c) the value of R.

माना ABC एक न्यूनकोण त्रिभुज है जिसका लम्बकेन्द्र H है। शीर्षो A, B, C से समाने वाली भुजा पर डाले गए लम्बों के पाद क्रमशः D, E, F है। ΔABC के परिवृत की त्रिज्या R है। दिया गया है कि $(AH)(BH)(CH) = 3$ तथा $(AH)^2 + (BH)^2 + (CH)^2 = 7$ है तो

(a) अनुपात $\dfrac{\prod \cos A}{\sum \cos^2 A}$, (b) गुणन $(HD)(HE)(HF)$ (c) R का मान ज्ञात कीजिये।

Ans. (a) $\dfrac{3}{14R}$, (b) $\dfrac{9}{8R^3}$, (c) $\dfrac{3}{2}$

Sol. $(AH)(BH)(CH) = 3$

$(2R)^3 [\cos A \cos B \cos C] = 3$ ...(1)

$AH^2 + BH^2 + CH^2 = 7$

$4R2 [\cos^2 A + \cos^2 B + \cos^2 C] = 7$ ...(2)

$\frac{(1)}{(2)}$

$\frac{8R^3}{4R^2}\frac{\pi \cos A}{\Sigma \cos^2 A} = \frac{3}{7}$

$\frac{\pi \cos A}{\Sigma \cos^2 A} = \frac{3}{14R}$

(b) (HD) (HE) (HF)

$8R^3 \cos^2 A \cos^2 B \cos^2 C$

$8R^3\left[\frac{3}{8R^3}\right]^2 \Rightarrow \frac{9}{8R^3}$

(C) $\cos^2 A + \cos^2 B + \cos^2 C$

$\cos^2 A - \sin^2 B + 1 + \cos^2 C$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$= \cos^2 A \cos^2 B - \sin^2 A \sin^2 B$

$= \cos^2 A(1 - \sin^2 B) - (1 - \cos^2 A)(\sin^2 B)$

$= \cos^2 A - \cos^2 A \sin^2 B - \sin^2 B + \sin^2 B$

$\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B$

$\Rightarrow 1 + \cos^2(A + B) + \cos(A + B) \cos(A - B)$

$\Rightarrow 1 + \cos(A + B)[\cos(A + B) + \cos(A - B)]$

$\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$

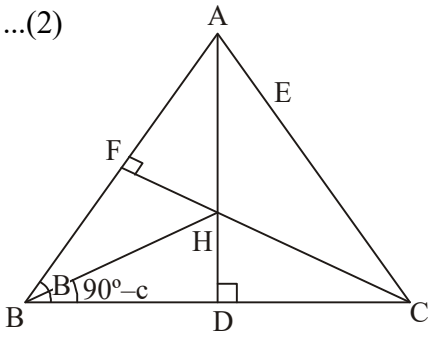
$\frac{7}{4R^2} = 1 - \frac{6}{8R^3}$

$BD = AB \cos B$

$BD = C \cos B$

$\cos(90 - C) = \frac{BD}{BH}$

$\sin C = \frac{C \cos B}{BH}$



$BH = \frac{C \cos B}{\sin C}$

$AH = \frac{b}{\sin B}\cos A$

$CH = \frac{a}{\sin a}\cos C$

$\tan(90 - C) = \frac{HD}{BD}$

$HD = BD \cot C$

$= \frac{C \cos^2 C}{\sin C}$

$HD = 2R\cos^2 C$

21. If Δ is the area of a triangle with side lengths a, b, c, then show that: $\Delta \le \frac{1}{4}\sqrt{(a+b+c)abc}$. Also show that equality occurs in the above inequality if and only if a = b = c.

यदि Δ ABC का क्षेत्रफल Δ है जिसकी भुजाऐं a, b, c है तो सिद्ध कीजिये की : $\Delta \le \frac{1}{4}\sqrt{(a+b+c)abc}$ तथा उपरोक्त असमिका में समानता a = b = c पर प्रकट होती है।

Ans.

Sol. To prove that $16\Delta^2 \le 2s \,.\, abc$

$\Rightarrow (s-a)(s-b)(s-c) \le \frac{abc}{8}$

Let $\begin{cases} s-a=x & \Rightarrow \quad x+y=c \\ s-b=y & \Rightarrow \quad y+z=a \\ s-c=z & \Rightarrow \quad z+x=b \end{cases}$

Applying AM $\ge$ GM

$x + y \ge 2\sqrt{xy}$ .....(i)

$y + z \ge 2\sqrt{yz}$ .....(ii)

$z + x \ge 2\sqrt{zx}$ .....(iii)

Multiplying (i), (ii) & (iii)

$(x + y)(y + z)(z + x) \ge 8(x\,y\,z)$

$\frac{abc}{8} \ge (s-a)(s-b)(s-c).$

22. If I_n is the area of n sided regular polygon inscribed in a circle of unit radius and O_n be the area of the polygon circumscribing the given circle, prove that $I_n = \frac{O_n}{2}\left(1+\sqrt{1-\left(\frac{2I_n}{n}\right)^2}\right)$.

यदि एक n भुजा वाले समबहुभुज जो एक इकाई त्रिज्या के वृत में समाहित है का क्षेत्रफल I_n है तथा वृत के परिगत समबहुभुज का क्षेत्रफल O_n है तो सिद्ध कीजिए।

$I_n = \frac{O_n}{2}\left(1+\sqrt{1-\left(\frac{2I_n}{n}\right)^2}\right)$

Ans.

Sol. Hint : put $I_n = \frac{n}{2}.\sin\left(\frac{2\pi}{n}\right)$

& put $O_n = n \tan\left(\frac{\pi}{n}\right)$.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 25

Consider a triangle ABC, where x, y, z are the length of perpendicular drawn from the vertices of the triangle to the opposite sides, a, b, c, respectively let the letters R, r, s, Δ denotes the circumradius, inradius semi-perimeter and area of the triangle respectively.

यदि त्रिभुज ABC में x, y, z शीर्षों से भुजाओं क्रमशः a, b, c पर डाले गये लम्बों की लम्बाईयाँ हैं तथा यदि R, r, s, Δ का सामान्य अर्थ हो तो निम्न को सरल करो।

23. If $\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = \frac{a^2+b^2+c^2}{k}$, then the value of k is :

यदि $\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = \frac{a^2+b^2+c^2}{k}$, हो तो k का मान होगा :

(A) R (B) s (C) 2R (D) $\frac{3}{2}$ R

Ans. C

Sol. $\frac{bc}{c} + \frac{cy}{a} + \frac{az}{b}$

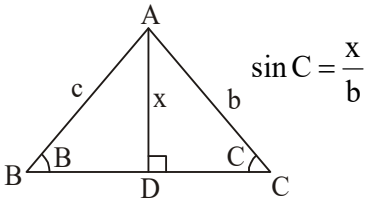
$b(\sin B) + c(\sin c) + a(\sin A)$

$\sin B = \frac{x}{c}$

Similarly

$\sin C = \frac{y}{a}$

$\sin A = \frac{z}{b} = \frac{c}{k}$



$\sin C = \frac{x}{b}$

$b \times \frac{a}{2R} + c \times \frac{c}{2R} + a \times \frac{a}{2R} \rightarrow \frac{a^2+b^2+c^2}{2R}$

24. If $\cot A + \cot B + \cot C = k\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right)$, then the value of k is :

यदि $\cot A + \cot B + \cot C = k\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$, हो तो k का मान होगा :

(A) R^2 (B) rR (C) Δ (D) $a^2 + b^2 + c^2$

Ans. C

Sol. $\cot A = \frac{\cos A}{\sin A} = \frac{2R\cos A}{a}$

$2R\left[\frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c}\right]$

$2R\left[\frac{b^2+c^2-a^2}{2abc}+\frac{a^2+c^2-b^2}{2abc}+\frac{a^2+b^2-c^2}{2abc}\right] \Rightarrow \frac{R}{abc}\left(a^2+b^2+c^2\right)$

$\sin B = \frac{x}{c} = \frac{b}{2R} \rightarrow x = \frac{bc}{2R}$

Similarly

$\frac{1}{y} = \frac{2R}{ac}$ $\frac{1}{x} = \frac{2R}{bc}$

$\frac{1}{z} = \frac{2R}{ab}$

$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2} = \frac{4R^2}{a^2b^2c^2}\left\{a^2+b^2+c^2\right\}$

$\frac{R}{abc}\left\{a^2+b^2+c^2\right\} = k\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$

$\frac{R}{abc}\left(a^2+b^2+c^2\right) = k\left(a^2+b^2+c^2\right)\frac{4R^2}{abc}$

$k = \frac{abc}{4R} \Rightarrow \Delta$

25. The value of $\frac{c\sin B + b\sin C}{x} + \frac{a\sin C + c\sin A}{y} + \frac{b\sin A + a\sin B}{z}$ is equal to :

$\frac{c\sin B + b\sin C}{x} + \frac{a\sin C + c\sin A}{y} + \frac{b\sin A + a\sin B}{z}$ का मान होगा :

(A) $\frac{R}{r}$ (B) $\frac{s}{R}$ (C) 2 (D) 6

Ans. D

Sol. $c \sin B = x$, $b \sin C = x$

adding $$\frac{c\sin B + b\sin C}{x} = \frac{2x}{x} = 2 \quad ...(i)$$

Similarly $$\frac{a\sin C + c\sin A}{y} = 2 \quad ...(ii)$$

$$\frac{b\sin A + a\sin B}{z} = 2 \quad ...(iii)$$

adding (i), (ii) & (iii), we get 6

Paragraph for Question Nos. 26 to 28

An altitude BD and a bisector BE are drawn in the triangle ABC from the vertex B. It is known that the length of side AC = 1, and the magnitudes of the angles BEC, ABD, ABE, BAC form an arithmetic progression.

त्रिभुज ABC के शीर्ष B से शीर्षलम्ब तथा कोर्णाद्धक क्रमशः BD तथा BE है यह दिया गया है कि AC = 1 तथा कोणों BEC, ABD, ABE, BAC के मान स.श्रेणी में हैं तो निम्न को सरल करो।

26. The area of circle circumscribing ΔABC is :

ΔABC के परिवृत्त का क्षेत्रफल होगा :

(A) $\frac{\pi}{8}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

Ans. B

27. Let 'O' be the circumcentre of ΔABC, the radius of circle inscribed in ΔBOC is :

यदि 'O' ΔABC का परिकेन्द्र है तो ΔBOC के अन्तःवृत्त की त्रिज्या होगी –

(A) $\frac{1}{8\sqrt{3}}$ (B) $\frac{1}{4\sqrt{3}}$ (C) $\frac{1}{2\sqrt{3}}$ (D) $\frac{1}{2}$

Ans. B

28. Let B' be the image of point B with respect to side AC of ΔABC, then the length BB' is equal to :

यदि B', AC भुजा के सापेक्ष B का प्रतिबिम्ब है तो इस ΔABC में BB' का मान होगा :

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{2}}{4}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Ans. D

Sol. Angles BEC, ABD, ABE and BAC are in A.P.

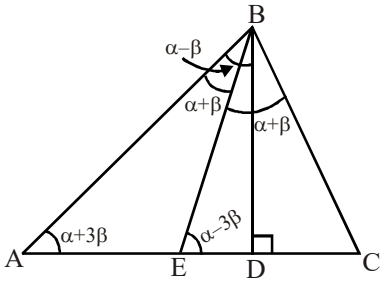
let BEC = $\alpha - 3\beta$

ABD = $\alpha - \beta$

ABE = $\alpha + \beta o$

and BAC = $\alpha + 3\beta$

now, $\alpha - 3\beta = (\alpha + 3\beta) + (\alpha + \beta)$ [using exterior angle theorem]



$\Rightarrow \alpha = -7\beta$

$\therefore \beta = -\frac{\pi}{24}, \alpha = \frac{7\pi}{24}$

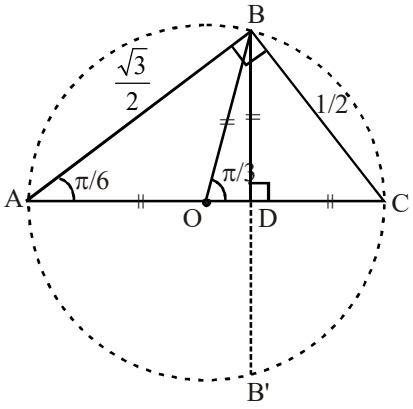
and From ΔABD

$$\alpha - \beta + \alpha + 3\beta = \frac{\pi}{2}$$

$$2\beta + 2\beta = \frac{\pi}{2} \quad \Rightarrow \alpha + \beta = \frac{\pi}{4}$$

$\therefore \angle B = 2(\alpha + \beta) = \frac{\pi}{2}, \angle A = \frac{\pi}{6}, \angle C = \frac{\pi}{3}$

$\Rightarrow$ ABC is 30°–90°–60° triangle



(i) Area of circle circumscribing $\Delta ABC = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$

(ii) ΔBOC is equilateral $\Rightarrow r = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4}\left(\frac{1}{2}\right)^2}{\frac{1}{2}\left(\frac{3}{2}\right)} = \frac{1}{4\sqrt{3}}$

(iii) $BD = OB \sin\frac{\pi}{3} = \frac{1}{2}\sin\frac{\pi}{3} = \frac{\sqrt{3}}{4}$

$\therefore BB' = 2BD = \frac{\sqrt{3}}{2}$

MATCH THE COLUMN TYPE QUESTION(S)

29. Match the following column :

Column-I	**Column-II**
(A) In a ΔABC, $(a + b + c)(b + c - a) = \lambda bc$, where $\lambda \in I$, then greatest value of λ is	(P) 3
(B) In a ΔABC, $\tan A + \tan B + \tan C = 9$. If $\tan^2 A + \tan^2 B + \tan^2 C = k$, then least value of k satisfying is	(Q) $9(3)^{1/3}$
(C) In a triangle ABC, then the joining the circumcentre to the incentre is parallel to BC, then value of $\cos B + \cos C$ is	(R) 1
(D) If ina ΔABC, $a = 5$, $b = 4$ and $\cos(A - B) = \frac{31}{32}$, then the find side c is equal to	(S) 6

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) ΔABC में, $(a + b + c)(b + c - a) = \lambda bc$, जहाँ $\lambda \in I$ तो λ का अधिकतम मान होगा	(P) 3
(B) ΔABC में , $\tan A + \tan B + \tan C = 9$. यदि $\tan^2 A + \tan^2 B + \tan^2 C = k$, हो तो इसको सन्तुष्ट करने वाला k का न्यूनतम मान होगा	(Q) $9(3)^{1/3}$
(C) त्रिभुज ABC में परिकेन्द्र तथा अन्तःकेन्द्र को जोड़ने वाली रेखा BC के समान्तर है तो $\cos B + \cos C$ का मान होगा	(R) 1
(D) यदि ΔABC में, $a = 5$, $b = 4$ तथा $\cos(A - B) = \frac{31}{32}$, तो भुजा c का मान होगा	(S) 6

Ans. (A) P ; (B) Q ; (C) R ; (D) S

Sol. (A) $(a + b + c)(b + c - a) = \lambda bc$

$(b + c)^2 - a^2 \Rightarrow b^2 + c^2 - a^2 = -2bc + \lambda bc$

$$\cos A = \frac{b^2 + c^2 - a^2}{bc} = (\lambda - 2)$$

$\lambda - 2 = 1 \rightarrow \lambda = 3$

(B) $A + B + C = \pi$

$\tan(A + B + C) = \tan \pi = 0$

$$\frac{\tan(A + B) + \tan C}{1 - \tan(A + B)\tan C} = 0$$

$\tan A + \tan B + (1 - \tan A \tan B)\tan C = 0$

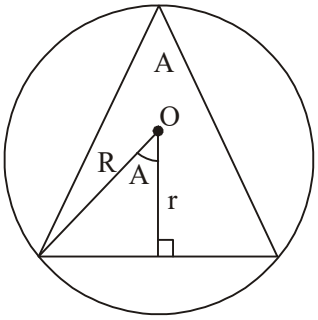
$\tan A + \tan B + \tan C = \tan A \tan B \tan C = 9$

Now applying AM & GM

$$\frac{\tan^2 A+\tan^2 B+\tan^2 C}{3} \geq \left(\tan^2 A \tan^2 B \tan^2 C\right)^{\frac{1}{3}}$$

$$\tan^2 A+\tan^2 B+\tan^2 C \geq 3(9)^{\frac{2}{3}}$$

(C) $\cos A=\frac{r}{R}$



$$R\cos A = r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$\cos A=\cos A+\cos B+\cos C-1$$

$$\cos B+\cos C=1$$

(D) a = 5, b = 4 and $\cos(A-B)=\frac{31}{32}$

$$\tan\left(\frac{A-B}{2}\right) \geq \frac{1}{\sqrt{63}} \qquad 2\cos^2\left(\frac{A-B}{2}\right)-1=\frac{31}{32}$$

$$\frac{a-b}{a+b}\cot\frac{C}{2} \qquad 2\cos^2\left(\frac{A-B}{2}\right)=\frac{31+32}{32}=\frac{63}{32}$$

$$\frac{1}{a}\cot\frac{C}{2}=\frac{1}{3\sqrt{7}} \qquad \cos^2\frac{A-B}{2}=\frac{63}{64} \rightarrow \cos\frac{A-B}{2}=\frac{\sqrt{63}}{8}$$

$$\cot\frac{C}{2}=\frac{3}{\sqrt{7}} \rightarrow \tan\frac{C}{2}=\frac{\sqrt{7}}{3} \rightarrow \cos\frac{C}{2}=\frac{3}{4}$$

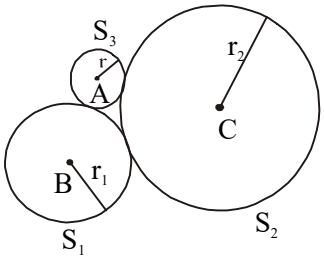
$$\cos C=\frac{a^2+b^2-c^2}{2ab} \qquad \cos C=2\cos^2\frac{C}{2}-1$$

$$\frac{1}{8}=\frac{25+16-c^2}{2\times4\times5} \qquad 2\times\frac{9}{16}-1$$

$$\cos C=\frac{1}{8}$$

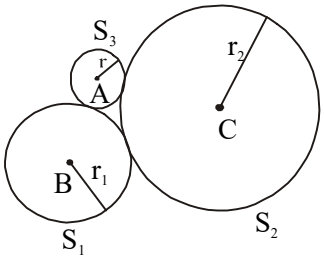
$c^2 = 36 \Rightarrow c = 6$

30. Consider two circles S_1 and S_2 having their centers at B and C with radii r_1 and r_2 respectively, touching each other externally. A circle S with radius r and centre at A touches both of them externally as shown in the figure :



Column –I	**Column –II**
(A) If ΔABC is right angled at A, then $\frac{r_2 + r}{r_2 - r}$ is	(P) $r \geq 1$
(B) If ΔABC is right angled at A and $r_1 . r_2 = 3 + 2\sqrt{2}$, then	(Q) $r_1 = r_2$
(C) If the common tangent of S_1 and S_2 at their point of contact passes through the centre of circle S, then	(R) $\frac{R}{4}$
(D) If $r_1 = r_2 = R$ and the circles S_1, S_2 and S have the same external common tangent then r is	(S) $\frac{r_1}{r}$

माना दो वृत S_1 तथा S_2 जिनका केन्द्र B तथा C पर है तथा इनकी त्रिज्याऐं क्रमशः r_1 तथा r_2 है तथा यह एक दूसरे को बाह्य स्पर्श करते है। एक r त्रिज्या वाला वृत S जिसका केन्द्र A पर है, दोनों वृतों को बाह्य स्पर्श करता है। (चित्रानुसार)

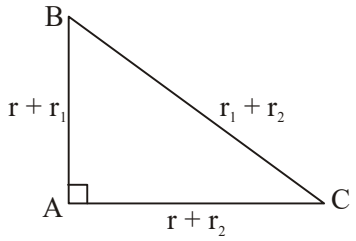


Column –I	**Column –II**
(A) यदि ΔABC एक समकोण त्रिभुज A पर है, तो $\frac{r_2 + r}{r_2 - r}$ का मान होगा	(P) $r \geq 1$
(B) यदि ΔABC एक समकोण त्रिभुज A पर है तथा $r_1 . r_2 = 3 + 2\sqrt{2}$, तो	(Q) $r_1 = r_2$
(C) यदि S_1 तथा S_2 के स्पर्श बिन्दु पर स्पर्श रेखा वृत S के केन्द्र गुजरती है, तो	(R) $\frac{R}{4}$
(D) यदि $r_1 = r_2 = R$ तथा वृतों S_1, S_2 तथा S की एक बाह्य उभयनिष्ठ स्पर्श रेखा है, तो r	(S) $\frac{r_1}{r}$

Ans. (A) S (B) P (C) Q (D) R

Sol. (A) if ΔABC is right angled at A

$(r_1 + r_2)^2 = (r + r_1)^2 + (r + r_2)^2$



$r_1^2 + r_2^2 + 2r_1r_2 = r^2 + r_1^2 + 2rr_1 + r^2 + r_2^2 + 2rr^2$

$r_1r_2 = r^2 + rr_1 + rr_2$

$r_1(r_2 - r) = r^2 + rr_2$

$r_1(r_2 - r) = r(r_2 + r)$

$$\frac{r_1}{r} = \frac{r_2 + r}{r_2 - r}$$

(B) if $r_1r_2 = 3 + 2\sqrt{2}$

$$= \left(\sqrt{2}\right)^2 + 1^2 + 2 \times 1 \times \sqrt{2}$$

$$r_1r_2 = \left(\sqrt{2} + 1\right)^2$$

$r^2 + r(r_1 + r_2) - r_1r_2 = 0$

$$r = \frac{-(r_1 + r_2) \pm \sqrt{(r_1 + r_2)^2 + 4(r_1r_2)}}{2}$$

$$r = \frac{-(r_1 + r_2) + \sqrt{(r_1 + r_2)^2 + 4r_1r_2}}{2}$$

$$\frac{r_1 + r_2}{2} \geq \sqrt{2} + 1 \Leftarrow \frac{r_1 + r_2}{2} \geq (r_1r_2)^{\frac{1}{2}}$$

$$r_1 + r_2 \geq 2\left(\sqrt{2} + 1\right)$$

$$r \geq \frac{-(r_1 + r_2) + \sqrt{4\left(\sqrt{2} + 1\right)^2 + 4\left(\sqrt{2} + r\right)^2}}{2}$$

$$r \geq \frac{2\sqrt{2}\left(\sqrt{2} + 1\right) - 2\left(\sqrt{2} + 1\right)}{2}$$

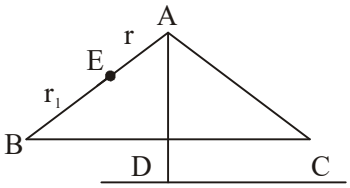
$$r \geq \left(\sqrt{2} - 1\right)\left(\sqrt{2} + 1\right)$$

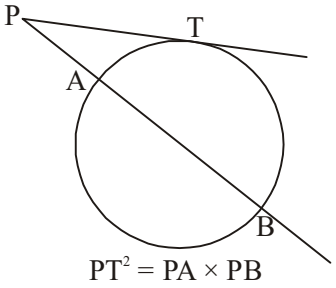
$r \geq 1$

(C) Circle properties

$AE \times AB = AD^2$

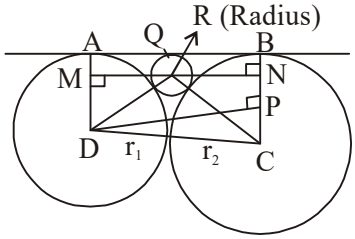
$AD^2 = rr_1 = rr_2 \Rightarrow r_1 = r_2$





$PT^2 = PA \times PB$

(D) let AB = d & QB = x $\quad \begin{bmatrix} \text{Here } \mu = r \\ \mu_1 = r_1 \\ \mu_2 = r_2 \end{bmatrix}$



$DP = d$

$PC = \mu_2 - \mu_1$

In ΔDPC $\qquad DC^2 = PC^2 + PD^2$

$(\mu_1 + \mu_2)^2 = (\mu_2 - \mu_1)^2 + d^2$

$d^2 = 4\mu_1\mu_2$

$d = \sqrt{\mu_1\mu_2}$

In ΔRMD $\qquad AQ = MR = d - x$

$(\mu_1 + \mu)^2 = (d - x)^2 + (\mu_1 - \mu)^2$ ...(1)

Similarly in ΔRNC

$(\mu^2 + \mu)^2 = (\mu^2 - \mu)^2 + x^2$

$x = 2\sqrt{\mu\mu_2}$

putting the value of x & d in equation (1)

$$4\mu_1\mu = 4\left(\sqrt{\mu_1\mu_2} - \sqrt{\mu\mu_2}\right)^2$$

$$\mu_1\mu = \mu_2\left(\sqrt{\mu_1} - \sqrt{\mu}\right)^2$$

$$\mu = \frac{\mu_1\mu_2}{\left(\sqrt{\mu_1} + \sqrt{\mu_2}\right)^2}$$

Now we have $\mu_1 = \mu_2 = R$ then $\mu = \frac{R^2}{4R} \Rightarrow \mu = \frac{R}{4}$

31. ABC is a triangle in which $\cos(A-B)=\frac{4}{5}$ and BC = 6, AC = 3. AD is the median through A, $\angle BAD = \alpha$, CL is perpendicular to AD :

Column –I	**Column –II**
(A) The value of $\sin\alpha$ is	(P) $\frac{3\sqrt{5}}{2}$
(B) Lenth of the median AD is	(Q) $\frac{1}{\sqrt{10}}$
(C) Radius of circumcircle of the triangle ABC is	(R) $3\sqrt{2}$
(D) The value of $\cot\angle ADC$ is	(S) 1

त्रिभुज ABC में $\cos(A-B)=\frac{4}{5}$ तथा BC = 6, AC = 3 है। AD माध्यिका जो कि A से गुजरती है, $\angle BAD = \alpha$, CL, AD के लम्बवत् है :

Column –I	**Column –II**
(A) $\sin\alpha$ का मान होगा	(P) $\frac{3\sqrt{5}}{2}$
(B) माध्यिका AD की लम्बाई होगी	(Q) $\frac{1}{\sqrt{10}}$
(C) त्रिभुज ABC के परिवृत की त्रिज्या होगी	(R) $3\sqrt{2}$
(D) $\cot\angle ADC$ का मान होगा	(S) 1

Ans. (A) Q (B) R (C) P(D) S

Sol. $\cos(A-B)=\frac{4}{5}$

$2\cos^2\left(\frac{A-B}{2}\right)-1=\frac{4}{5}$

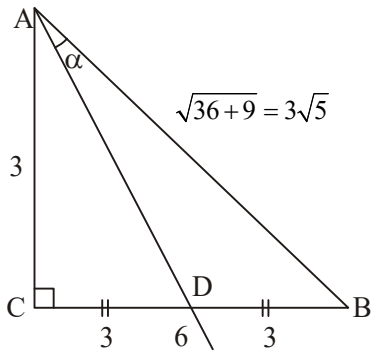
$2\cos^2\left(\frac{A-B}{2}\right)=\frac{9}{5}$

$\cos^2\left(\frac{A-B}{2}\right)=\frac{9}{10}$

$\cos\left(\frac{A-B}{2}\right)=\frac{3}{\sqrt{10}}\Rightarrow\tan\left(\frac{A-B}{2}\right)=\frac{1}{3}$

a = BC = 6

b = 3



we know that

$$\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}\cot\frac{c}{2}$$

$$\frac{1}{3}=\frac{6-3}{6+3}\cot\frac{c}{2}$$

$$\frac{1}{3}=\frac{3}{9}\cot\frac{c}{2}\Rightarrow\cot\frac{c}{2}=1$$

$$c=\frac{\pi}{2}$$

$$AD=3\sqrt{2}\qquad\qquad \sin B=\frac{3}{3\sqrt{5}}=\frac{1}{\sqrt{5}}$$

$$\frac{3}{\sin\alpha}=\frac{3\sqrt{2}}{1}\times\sqrt{5}\Rightarrow\sin\alpha=\frac{1}{\sqrt{10}}$$

$$2R=\frac{b}{\sin B}=\frac{3}{\frac{1}{\sqrt{5}}}=3\sqrt{5}\rightarrow R=\frac{3\sqrt{5}}{2}$$

ΔABC $\cot D=\frac{3}{3}\Rightarrow\cot D=1$